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Project Title: PlantCare: Smart Garden Monitoring and Management

Project Objective:

The Smart Garden Monitoring System is designed to store and manage essential data about small home plants and flowers. It focuses on soil moisture and sunlight exposure levels. The primary objective is to develop a structured, accessible database that records and analyzes environmental readings. This helps users maintain their gardens more conveniently. This eco-friendly system promotes sustainable gardening practices and inspires individuals to take an active interest in nurturing plants and contributing to a greener environment. Furthermore, the system is designed with scalability in mind, allowing future integration of additional features such as IoT sensors or automation, thereby expanding its practical applications.

Functionalities:

1. Pre-Stored Plant Data

- **Reference Database:** Provide users with previously stored soil moisture and sunlight data for a variety of houseplants.

- **Optimal Care Guidelines:** Include recommended daily watering (kg/liter/gallons) and sunlight exposure duration for each plant.

- **User Reference:** Help users understand standard care requirements before tracking their own plant conditions.

2. Current Plant Status Tracking

- **Daily Input:** Allow users to record how much water was given and how long each plant was exposed to sunlight.

- **Real-Time Monitoring:** Track daily plant care activities to ensure consistency with recommended standards.

- **Individual Plant Records:** Maintain ongoing data for each user's plants to monitor patterns and progress.

3. Data Analysis, Feedback, and User Engagement

- **Comparative Analysis:** Compare the user's daily plant data with pre-stored reference values to detect deviations from optimal care.

- **Interactive Reports & Alerts:** Provide daily summaries with actionable insights, notifications for watering or sunlight adjustments, and personalized care suggestions.

- **Engaging Experience:** Promote convenient, interactive plant care that motivates users to maintain healthy plants consistently.

4. Monthly Summary Reports

- **Monthly Trends & Goal Tracking:** Show watering, sunlight, and plant health trends while tracking personalized care goals.

- **Interactive & Sustainable:** Provide visualizations that promote consistent engagement and eco-friendly gardening habits.

Expected Outcomes:

- **Enhanced Plant Care Accessibility:** A user-friendly platform that helps users easily track and manage the care of their houseplants.
- **Real-Time Monitoring:** Daily tracking of watering and sunlight exposure ensures optimal plant health.
- **Structured Data Management:** A secure, organized system that stores pre-set reference data and user-input plant status, supporting daily comparisons and growth tracking.
- **Interactive Engagement:** Provides interactive reports, visualizations, and goal tracking to motivate users and encourage consistent plant care.
- **Sustainable System:** Promotes eco-friendly habits by helping users monitor their plants, contributing to the environment's well-being.
- **Scalability & Future Enhancements:** A modular system design capable of incorporating additional plant species, advanced analytics, or integration with smart gardening devices.